

First Level of Security Certification report Systeme d'Arme Furtif Enclavé

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2 INFORMATIONS

Product Name	Système d’Arme Furtif Enclavé (SAFE)
This document reference	SSTIC_2026_SAFE_v1.0.0
Product Category	Identification, Authentication and Access Control
Evaluation reference	First Level of Security Certification (CSPN in French).
Commissionner	Aegis Arms Technologies
Developer	Aegis Arms Technologies
Evaluation Center	KaKaWuet – Rennes
Evaluated Security Functions	Identification, Authentication and Access Control

3 PRODUCT

3.1 PRODUCT PRESENTATION

The evaluated product is SAFE (Système d’Arme Furtif Enclavé), Version 1.0.0, developed by Aegis Arms Technologies. The product is presented as a software access control system for the missile launch platform from a global directory to an entity.

3.2 PRODUCT DESCRIPTION

The Target Of Evaluation [TOE] defines the evaluated product, its security features, and its operational environment.

3.2.1 Product Category

The product falls under the category of "Identification, Authentication and Access Control"

3.2.2 Product Identification

The analyzed version of the product is version 1.0.0

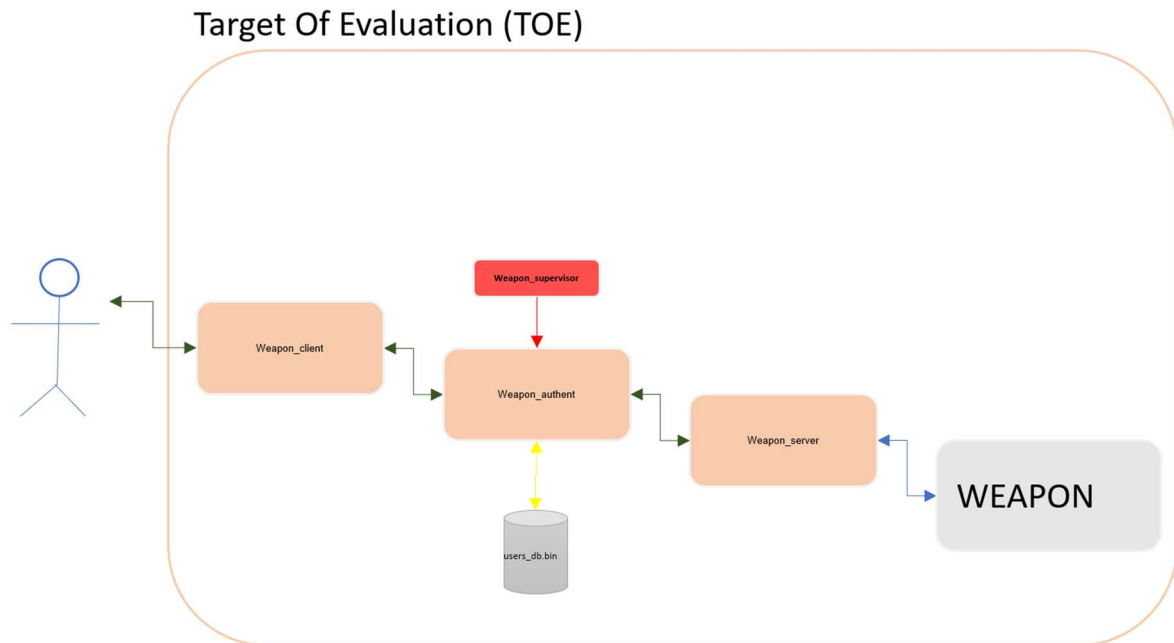
3.2.3 Security Functions

The product's security functions are as follows:

- User authentication and access control

- Data confidentiality and integrity protection
- Cryptographic functions and random number generator

3.2.4 Evaluated Configuration



3.2.5 Target of Evaluation: Summary

3.2.5.1 Sensitive Assets

The sensitive assets are as follows :

- User database of the SI (shared with the entire administration)
- Access to weapon (especially arm/disarm actions)

3.2.5.2 Threats

3.2.5.2.1 Threats on the database

An unauthorized user gains read and/or write access to the database.

3.2.5.2.2 Theats on the argming mechanism

The security functions used to protect sensitive assets are as follows:

3.2.5.3 Security Functions

The security functions used to protect sensitive assets are defined in the following table.

Security Function ID	Security Function	Description
[SF_Access_Rights_DB]	Access Rights to Database	The database is protected for read/write access using the Linux access rights defined during installation.
[SF_Access_Rights_ArmingDevice]	Access rights to arming device	Access to arming device is protected using Role Based Access Control.
[SF_ROP_Exploitation_Protection]	Exploitation protection	Custom security mechanism is enforced to prevent ROP exploitation on the authentication server.

3.2.5.4 Security Assumptions

Security Assumptions ID	Security Assumptions	Description
[SA_No_Physical_Access]	No Physical Access	There is no physical access to the machine for an attacker. Users having physical access are trusted.
[SA_No_Network_Access]	No Network Access	There is no network access to the machine for an attacker, a diode is used to transfert updates. Users having network access are trusted.
[SA_Compliant_Installation]	Compliant Installation	Installation has been performed accordingly to the documentation, and all security measures are enforced.

4 EVALUATION

4.1 EVALUTION REFERENCES

The evaluation was carried out in accordance with the First Level of Security Certification (CSPN in French).

4.2 EVALUATION WORK

The evaluation work was carried out based on the need for security, sensitive assests, threats, users, and security functions define in the TOE.

4.2.1 Installation

Installation was performed using the documentation without any problems.

4.2.2 Documentation Analysis

The documentation is clear and precise. It allows the product to be used without accidentally degrading security.

4.2.3 Source code review

No source code review was carried out in relation to the security target. After all, the target is only accessible to trusted users.

4.2.4 Security Function Compliance Analysis

Security Function ID	Security Function	Comment	Result
[SF_Access_Rights_DB]	Access Rights to Database	Access rights to database are correctly set by installer. Trying to read the database file as a simple user fails.	SUCCESS
[SF_Access_Rights_ArmingDevice]	Access rights to arming device	Access rights to arming device is correctly enforced. Trying to perform action "FIRE" or "DISARM" is blocked by the authentication	SUCCESS

		server if the user doesn't have the capability.	SUCCESS
[SF_ROP_Exploitation_Protection]	Exploitation protection	ROP exploitation is prevented by custom mechanism.	

4.2.5 Vulnerability analysis

4.2.5.1 Known Vulnerabilities

There are no known vulnerabilities on the Target of Evaluation.

However, we can list the following highly dangerous vulnerabilities affecting the Linux system, on which the TOE is executed.

[illegible]

4.2.5.2 *List of Vulnerabilities Discovered During Evaluation and Expert Opinion*

Vulnerabilities have been discovered and identified, but they are not exploitable in the context defined by the security target. Vulnerability exploitation is prevented by :

- [SA_No_Physical_Access]
- [SA_No_Network_Access]
- [SA_Compliant_Installation]
- [SF_ROP_Exploitation_Protection]

4.2.6 Ease of use analysis

4.2.6.1 *Cases where Security is not fully enforced*

The security assumption [SA_Compliant_Installation] indicates that product is correctly installed and follow security guidelines. Especially, not having the supervisor running may reduce security drastically.

That said, an on site audit have been performed and showed that arming device was correctly installed and deployed.

4.2.6.2 *Expert analysis on ease of use*

The product is well documented, and the Python client interface can be easily customized.

4.3 ANALYSIS OF THE RESISTANCE OF CRYPTOGRAPHIC MECHANISMS

The only cryptographic mechanism is the user password mechanism. Except for id/pw couple "SSTIC_USER:DefaultPassword", every other passwords are random 255-character strings, which are then hashed using the SHA256 algorithm.

Moreover, the database and users are all changed regularly, at each restructuring (about every 3 months).

4.4 ANALYSIS OF THE RANDOM NUMBER GENERATOR

The analysis of the random number generator did not reveal any non-complicance with the ANSSI cryptographic reference.